**E-COMMERCE SALES DATA ANALYTICS**

*Project Documentation: Data Cleaning, Analysis and Visualization with Excel*

## 1. Project Objective

The objective of this project is to clean and prepare an e-commerce dataset, calculate sales measures, combine related data using Excel lookup functions, summarize performance with PivotTables, and present results through charts and a dashboard.

## 2. Dataset Description

The workbook contains related tables for customers, products, stores, sales transactions, stock and payment information. The Sales_Fact sheet contains transaction-level fields used to calculate sales and build the dashboard.

## 3. Column Description

| Column | Description |
|---|---|
| Sales_ID | Unique identifier for a sales transaction |
| Order_Date | Date associated with the order |
| Customer_ID | Identifier linking a transaction to customer details |
| Product_ID | Identifier linking a transaction to product details |
| Store_ID | Identifier of the store associated with the sale |
| Quantity | Number of units sold |
| Unit_Price | Price per unit |
| Discount | Discount rate applied to the transaction |
| Payment_Type | Payment method used by the customer |
| Total_Amount | Calculated transaction amount after discount |
| Product_Name / Customer_Name | Lookup results that make IDs easier to interpret |

## 4. Data Cleaning and Transformation

- Reviewed the Customer_Dim, Product_Dim, Store_Dim and Sales_Fact sheets.
- Used Excel's Remove Duplicates tool to check for duplicate records.
- Used text-cleaning functions such as TRIM, CLEAN and PROPER in the cleaning workflow.
- Checked missing values. Three missing Discount values were filled with the average Discount value as an estimate.
- Prepared Quantity, Unit_Price, Discount and Total_Amount fields for analysis.
- Used XLOOKUP and VLOOKUP to bring product and customer names into the sales table.

Note: Average-based values used to fill blanks are estimates, not the original recorded values.

## 5. Formulas Used

**Total sales:** =SUM(J2:J2001)
Adds values in Total_Amount.

**Payment-type count:** =COUNTIF(I2:I2001,"Credit Card")
Counts transactions for a payment method; the criterion can be changed to PayPal or COD.

**Average unit price:** =AVERAGE(G2:G2001)
Calculates the mean Unit_Price.

**Price category:** =IF(G2>500,"High","Low")
Classifies unit prices above 500 as High and the rest as Low.

**Formatted order date:** =TEXT(B2,"dd-mm-yyyy")
Displays the order date as day-month-year.

**Product name lookup:** =XLOOKUP(D2,Product_Dim!$A$2:$A$101,Product_Dim!$B$2:$B$101,"Not Found")
Returns the product name matching Product_ID.

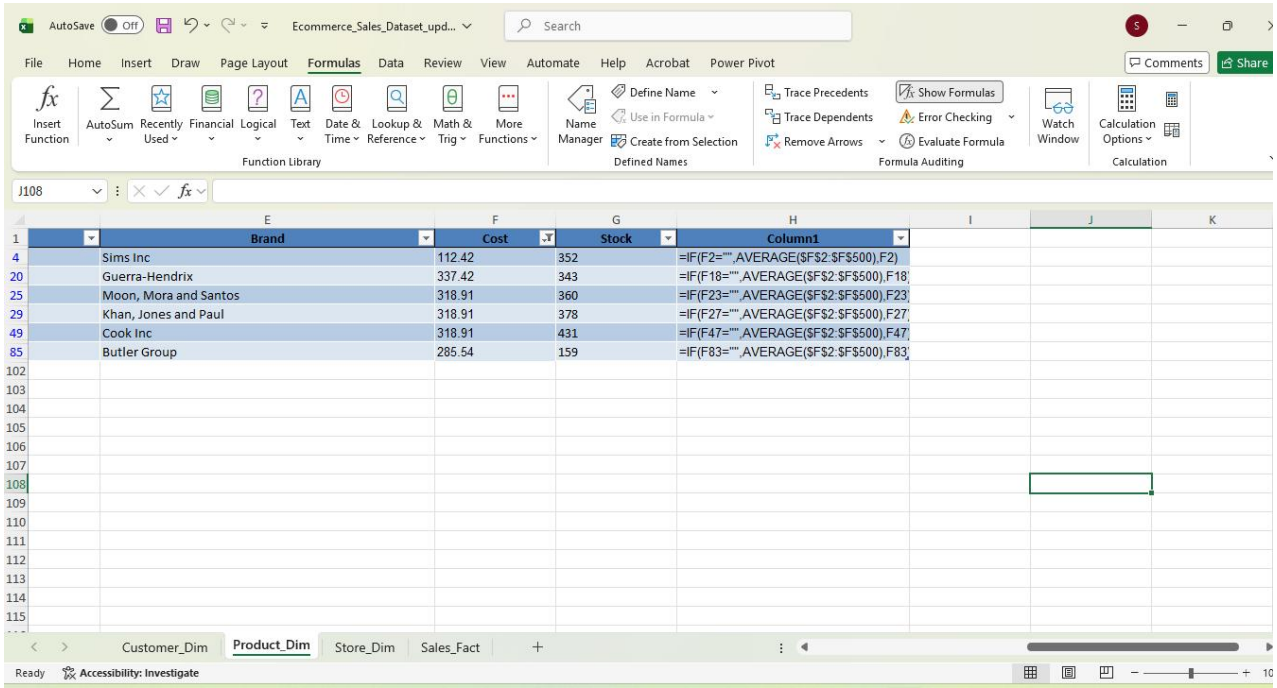
**Customer name lookup:** =VLOOKUP(C2,Customer_Dim!$A$2:$B$2001,2,FALSE)
Looks up customer details using Customer_ID.

**Total amount:** =F2*G2*(1-H2)
Calculates quantity multiplied by unit price after discount.

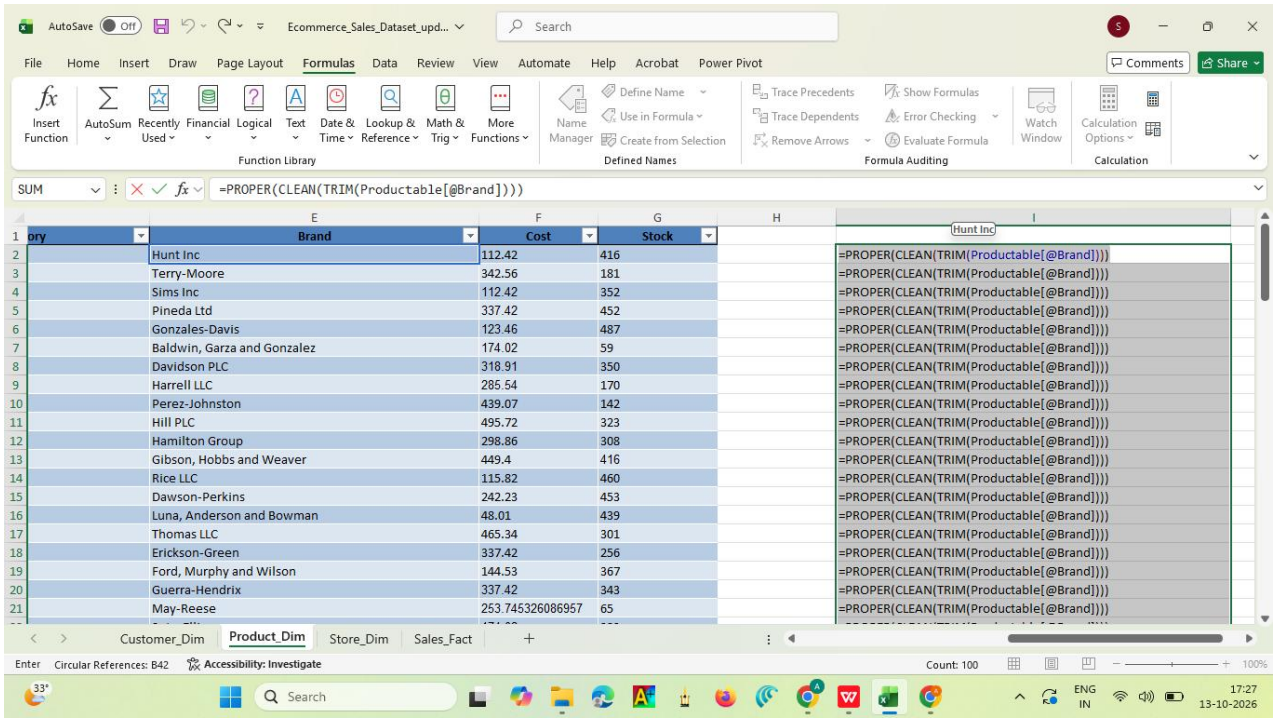
## 6. Screenshots of Cleaning and Formulas

The following screenshots document the Excel work completed during data cleaning, formula use and lookup preparation.

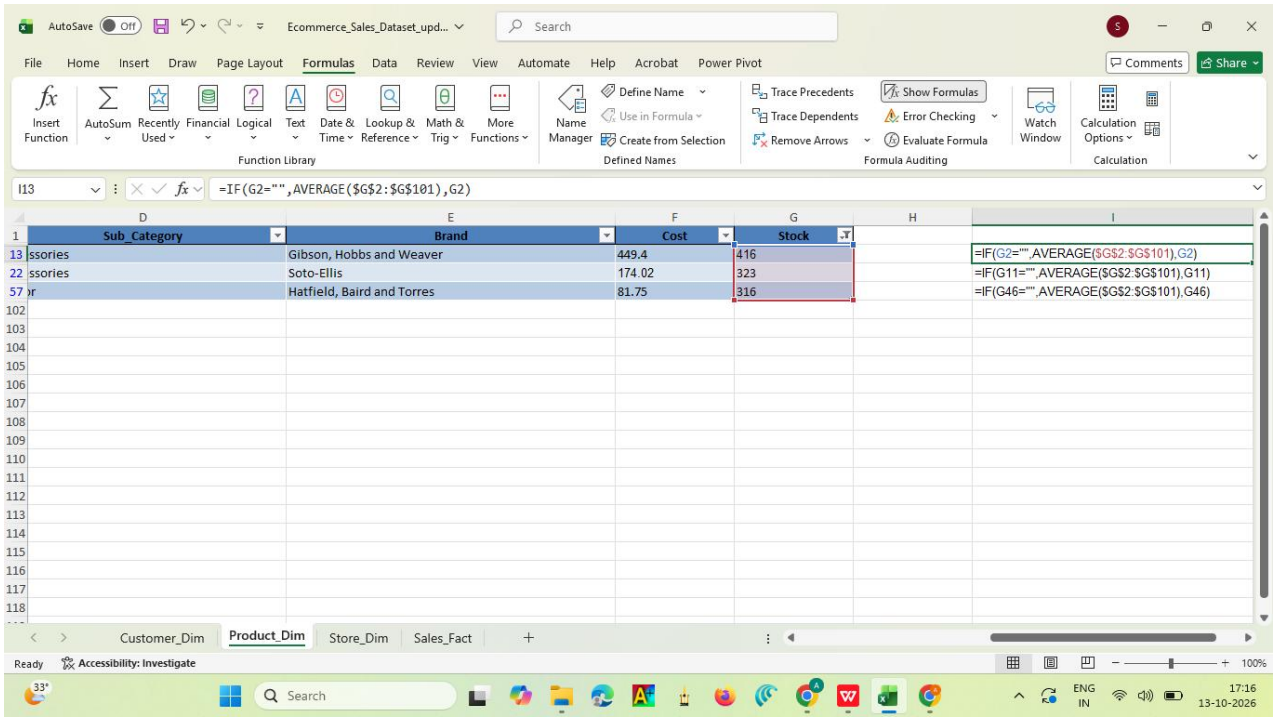
### Customer data and duplicate-removal step



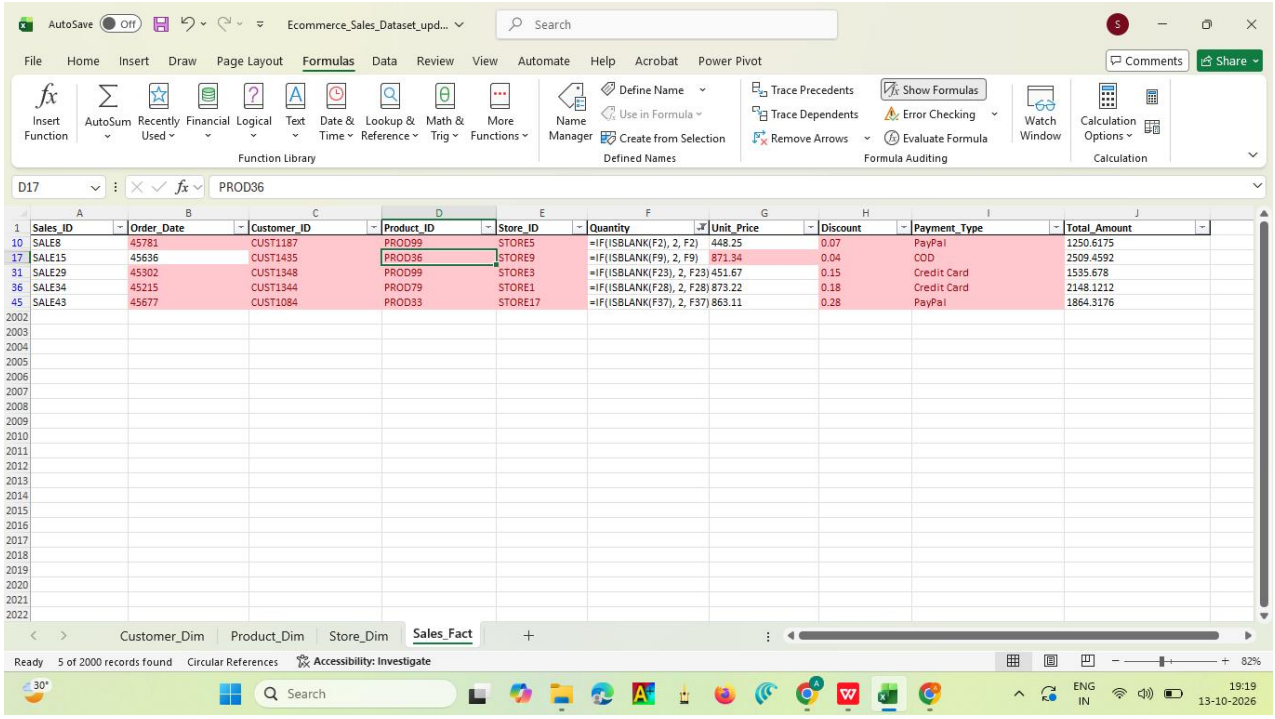

### Product dimension data cleaning and formula



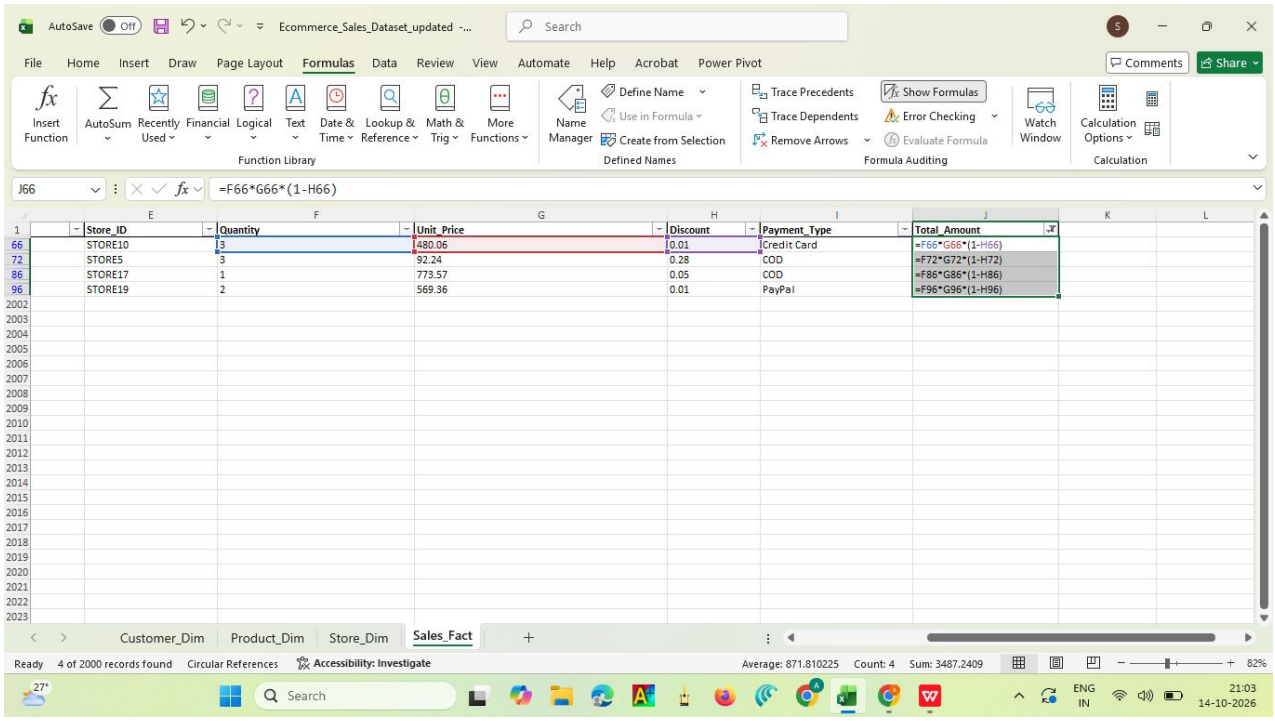

## Product dimension missing-value handling



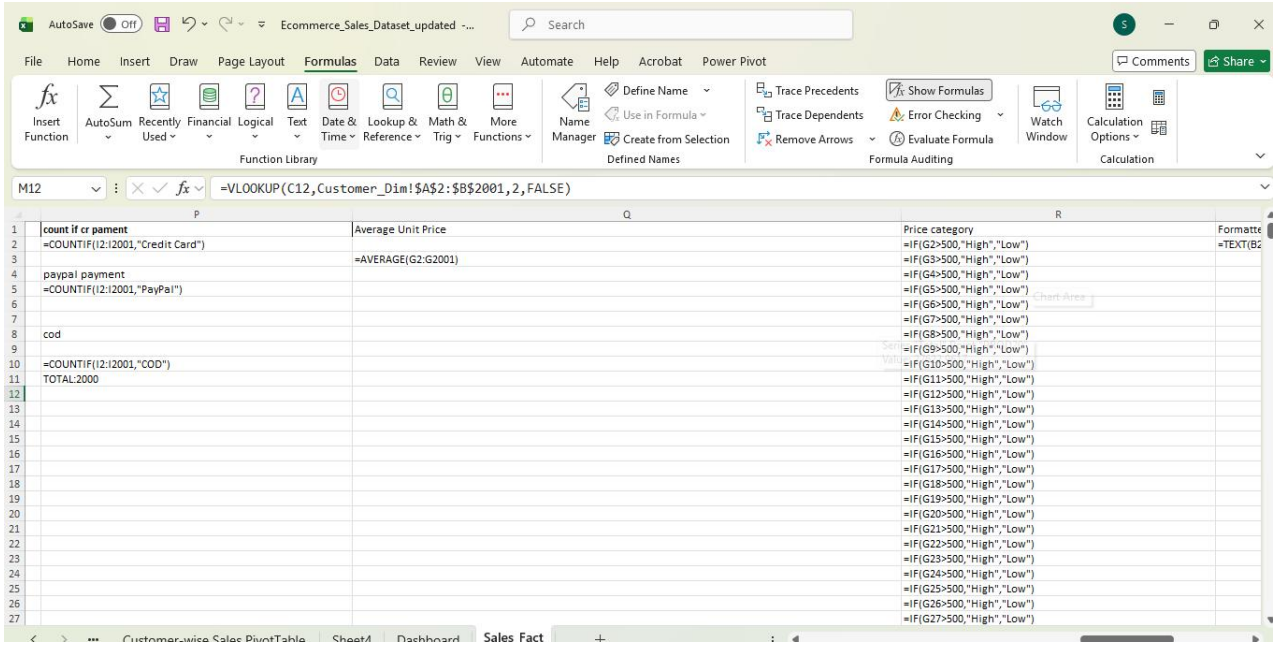

## Product dimension cleaning formula examples



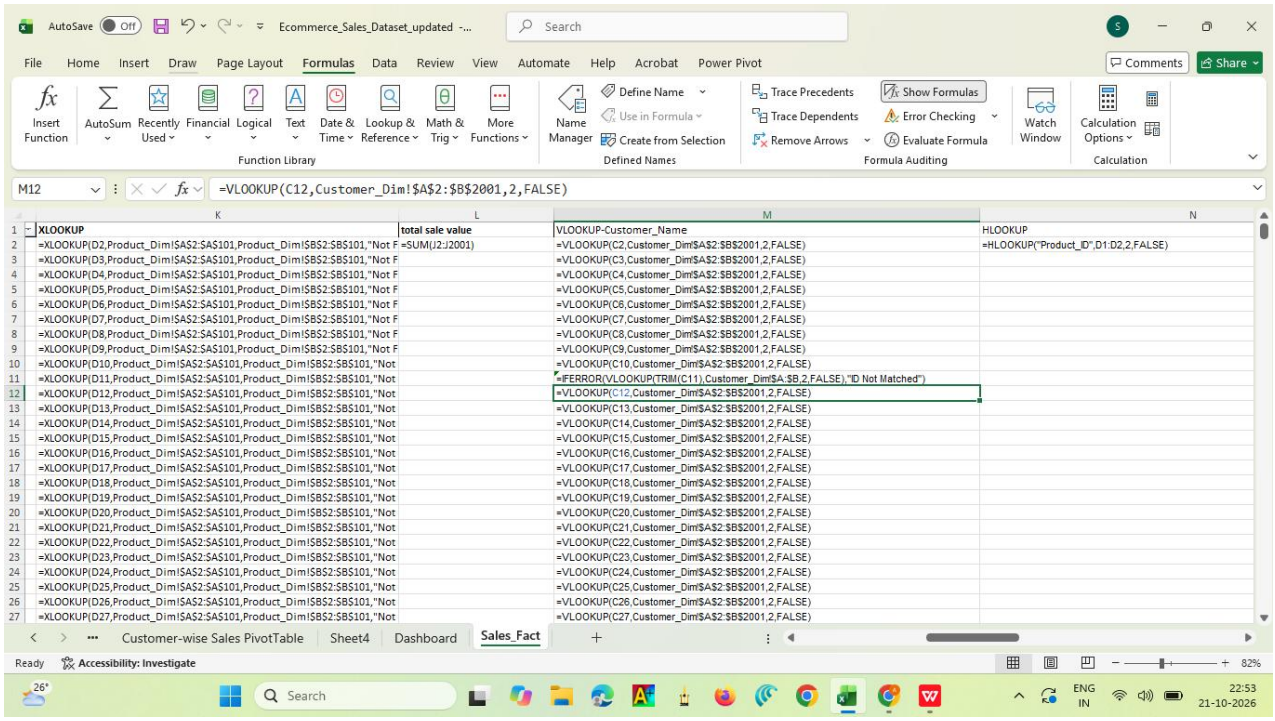

## Sales Fact data and total amount formula




## Sales Fact lookup formulas



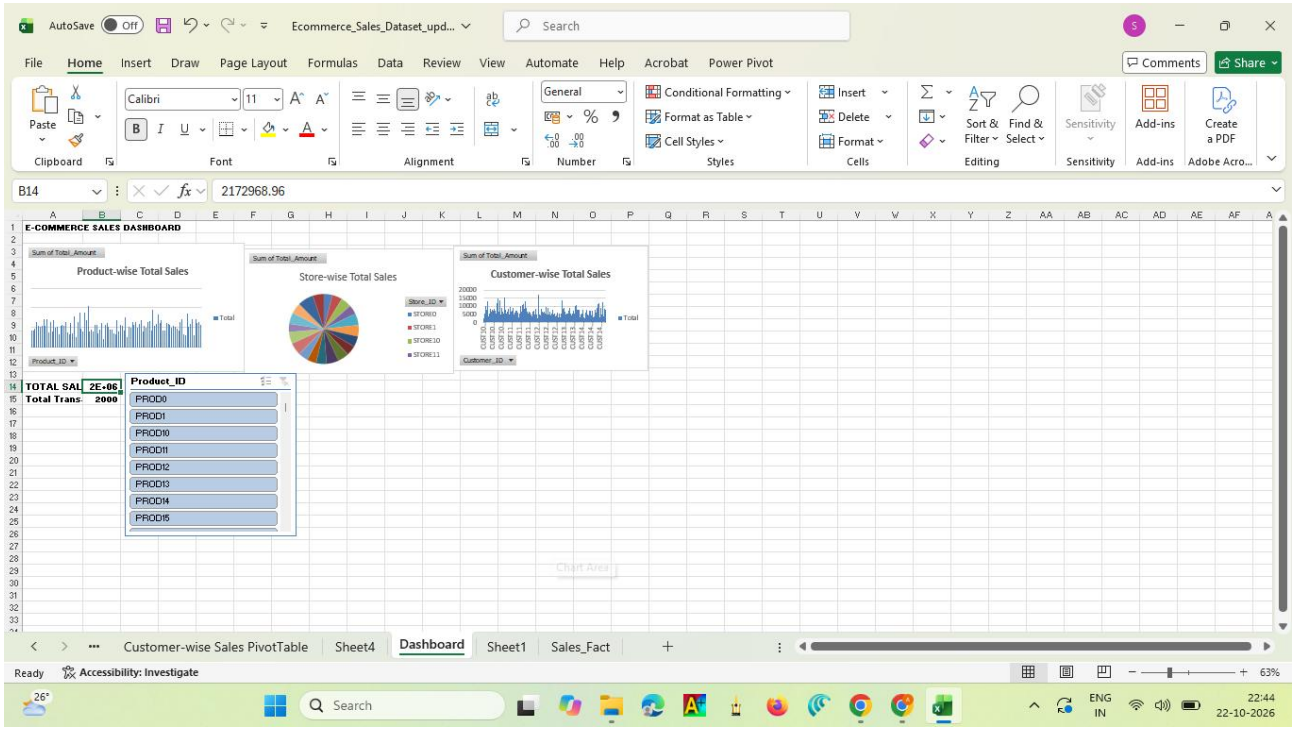

## Sales Fact lookup formulas and summary calculations



## 7. Visualization and Insights

PivotTables were created to summarize total sales by payment type, product, store and customer. The dashboard combines product-wise, store-wise and customer-wise sales charts with total sales and transaction count.

### Dashboard Screenshot




## 8. Analysis Types

### Descriptive Analysis

The dataset contains 2,000 sales transactions. Total sales are approximately 2,172,968.96. Average Unit_Price is approximately 513.60, median is 522.43, mode is 871.34 and standard deviation is approximately 282.33.

### Diagnostic Analysis

Payment-type summaries show PayPal sales of 756,754.57, Credit Card sales of 713,754.97 and COD sales of 702,459.42. PayPal has the highest total sales among these payment types.

### Predictive Analysis

The workbook summarizes historical transactions and does not include a validated forecasting model. Historical patterns could be used as a starting point for forecasting, but no forecast value is claimed here.

### Prescriptive Analysis

Use the dashboard to compare product, store and customer performance. Investigate high-performing products and payment methods, review store-level differences and continue checking missing values or unmatched IDs.

## 9. Four Key Insights as Storytelling

1. Overall performance: 2,000 recorded transactions generated approximately 2.17 million in total sales.

2. Payment preference: PayPal recorded the highest total sales among the listed payment types, followed by Credit Card and COD.
3. Price distribution: Average Unit_Price is about 513.60 and the median is 522.43, giving a view of typical transaction pricing.
4. Decision support: Product-wise, store-wise and customer-wise charts help identify performance differences and areas for further investigation.

## 10. Conclusion

This project demonstrates an Excel workflow for data cleaning, formula-based calculations, lookup functions, PivotTable summaries and dashboard visualization. Before submission, verify that lookup formulas return matched names and save the workbook with the final dashboard filters cleared.